Interferon-Gamma Receptor Ligation Triggers JAK1 Activation and STAT1 Expression in Macrophages

Authors: Zhi-Wei Wang, MD, Beatriz Morales, PhD
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Page 1 — Abstract & Introduction

Interferon-gamma (IFN-gamma) orchestrates host antimicrobial defense and macrophage polarization toward the M1 phenotype.

Page 2 — Receptor Signaling

Ligation of IFN-gamma to its receptor induces trans-phosphorylation and activates JAK1 kinase at the cell surface.

Page 3 — STAT1 Phosphorylation

Active JAK1 phosphorylates STAT1 at Tyr701, driving STAT1 homodimerization and nuclear entry.

Page 4 — Gene Expression

Nuclear STAT1 binds GAS promoter elements to upregulate IRF1 expression in activated macrophages.

Page 5 — Discussion

In conclusion, IFN-gamma drives IRF1 expression via sequential JAK1 and STAT1 activation.